
Bérince S. R. Hounsouvo

Ph. D Candidate in Forest sciences, Specializing in Tree-Fungi-Soil Interactions | [Université Laval](#)
1030 Av. de la Médecine, Québec, QC G1V 0A6, Canada
bshou@ulaval.ca | <https://www.linkedin.com/in/bérince-hounsouvo> | Updated: April 21, 2026

Education

Ph.D. in Forest Sciences

2023 - 2026

Department of wood and forest sciences, Université Laval, Québec Canada

Thesis: [Selection of boreal woody species with lateral root architecture for technosol-based revegetation of acid mine drainage sites](#)

Supervisor: [Prof. Damase P. Khasa](#)

Co supervisor: [Prof. Antoine Karam](#)

Master's degree, Forest Sciences and Techniques

2018 - 2020

Faculty of Agronomy, University of Parakou, (Benin)

Thesis: [Effect of ectomycorrhizal inoculation and organic and mineral amendments on juvenile growth of nursery plants of Afzelia africana, Pterocarpus erinaceus & Khaya senegalensis.](#)

Supervisor: [Prof. Christine Ouinsavi](#)

Bachelor's degree, Natural Resource Management

2014 - 2017

Faculty of Agronomy, University of Parakou, (Benin)

Thesis: [Optimal sampling technique for plot-based biodiversity measurement in the Sudanian zone: the case of square and circular plots](#)

Supervisor: [Prof. Nourou S. Yorou](#)

Additional Training

[6] [Hierarchical models and Bayesian inference for the natural sciences](#), Winter 2026, Université Laval, QC, Canada

[5] [Big data processing and functional predictions in Biology - SymbioSol workshop 2025](#), Quebec, Canada.

[4] [NSERC CREATE Program on Nature-Based Solutions for Ecosystem Restoration - NASER Summer School 2025 \[in person\]](#) . Université Laval, Quebec, Canada. From April 28 to May 09, 2025

[3] [Programming with R for data analysis \[in person\]](#). Université Laval, Québec, Canada.

[2] [Introduction to Unix Shell \[online, UNIX101\]](#). Calcul Québec, Québec, Canada. February 2, 2024

[1] [Data analysis in bioinformatics: exploring computational biology \[in person\]](#). Université Laval, Quebec, Canada. From April 28 to May 09, 2025

Research Interests

- Forest biology
- Mycorrhizal symbioses
- Forest ecophysiology
- Restoration ecology
- Urban forestry

Academic and Research Experiences

Research Assistant, Forest Biology

2024 - present

Department of wood and forest sciences, Université Laval, Québec Canada.

- Assist a PhD candidate in designing a forest field experiment
- Participate in ECM (ectomycorrhizal) seedling inoculation
- Manage fungal strain collection

Doctoral Researcher, Forest biology

2023 - present

Department of wood and forest sciences, Université Laval, Québec Canada.

- Develop literature reviews on our scientific knowledge of tree-soil and tree-fungi interactions.
- Design and carry out advanced experimental protocols in forestry and restoration practices.
- Build and run advanced statistical models using R and JAGS.

Research Associate, Forest Biology

2025 - present

Forest Studies and Research Laboratory, University of Parakou, Parakou (Benin)

- Design and conduct forest experimentation projects
- Conduct analyses using linear mixed-effects models in R
- Write scientific and field mission reports

Teaching Assistant, Forest Biology

Winter 2025

Department of soils and agro-food engineering, Université Laval, Québec Canada

- Co-supervise a master's student in the design and implementation of a forest experiment
- Provide scientific guidance on experimental design and field data collection

Research Assistant, Tropical Mycology

2017 - 2018

Research Unit in Tropical Mycology and Plant-Fungi-Soil Interactions, Parakou (Benin)

- Conduct surveys of macro-fungi in open forests and savannas
- Evaluate fungal diversity indices (α and β diversity) using R software
- Manage fungi specimens in the Mycological Herbarium collection at the University of Parakou

Publications

[5] **Hounsouvo BSR** et al., Using mycorrhizal inocula for tree in Africa: an integrative conceptual framework and research roadmap (in preparation for Biological reviews).

[4] **Hounsouvo BSR**, Khasa DP, Karam A and Lamhamedi MS. Tree root development and architecture at post-mining sites: A comprehensive systematic review (in preparation for STOTEN).

[3] **Hounsouvo BSR**, Orou Sourou Z, Ayena JIK, and Dramani R. Socio-Agronomic Determinants of Farmland Access in Vegetable Production Systems in Northern Benin: Evidence from Bassila. Agriculture and Human Values (submitted). <https://doi.org/10.21203/rs.3.rs-6614725/v1>

[2] **Hounsouvo BSR**, Houehanou TD, Gouwakinnou GN, Hadonou CJ, and Ouinsavi CAIN. (2022). Comparative Effect of Ectomycorrhizal Inoculation, Mineral and Organic Amendments and their Interactions on Juvenile Growth of *Azadirachta indica* in the Nursery. Journal of Tropical Plant Physiology, 14(1), 1-12. 022):1-12 <https://doi.org/10.56999/jtpp.2022.14.1.18>

[1] Habakaramo MP, Tchan KI, Hegbe ADMT, Abohombou G, **Hounsouvo BSR**, Tchemagnon O, Dramani R, and Yorou NS (2022). Sampling techniques for the optimal measurements of macromycetes diversity in the Soudano-Guinean ecozone (West Africa). International Journal of Biodiversity and Conservation, 14(3), 128-138. <https://doi.org/10.5897/IJBC2020.1404>

Scientific Communications

[4] [Root System Strategies of *Picea mariana* and *Larix laricina* for Revegetation of Acid Mine Drainage Sites in Quebec](#). Journée étudiante de Centre SÈVE, Quebec, Canada. [Poster]

[3] [Variation génétique de la croissance racinaire chez les plants d'épinette noire et de mélèze laricin](#). Colloque forestier 2025. L'office des producteurs de plants forestiers du Québec (OPPFQ) Québec, Canada. [Oral presentation]

[2] [Genetic Control and Repeatability of Root Growth in Seedlings of EPN and MEL, Two Species with Contrasting Strategies Along the Root Economics Spectrum](#). Interlab 2025. Institute of Integrative Biology and Systems, Université Laval, Quebec, Canada. [Oral presentation]

[1] [Root System Strategies of Picea mariana and Larix laricina for Revegetation of Acid Mine Drainage Sites in Quebec](#). Journée étudiante de l'IBIS (15e ed), Institute of Integrative Biology and Systems, Université Laval, Quebec, Canada. [Poster]

Awards - Grants - Fellowships (\$84,650)

Doctoral degree - \$81,400

[9] J. André Fortin grant in Forest Mycology and Forest Soil Biology: Influence of root morphology on the mycorrhizal community of boreal woody species (2026) - \$2500.

[8] Doctoral thesis writing grant, Faculty of Graduate and Postdoctoral Studies, Université Laval (2026) - \$5000.

[7] Best PhD progress award (37RAPF), Faculty of Graduate and Postdoctoral Studies, Université Laval (2025) - \$3700.

[6] Participation Grant for the Student Day 2025, Centre SÈVE: Root System Strategies of Picea mariana and Larix laricina for Revegetation of Acid Mine Drainage Sites in Quebec (2025) - \$1000

[5] Doctoral project completion Award (37EXAD), Faculty of Graduate and Postdoctoral Studies, Université Laval (2020) - \$6200

[4] Award for doctoral project preparation, Faculty of Graduate and Postdoctoral Studies, Université Laval (2024) - \$3000

[3] Doctoral fellowship, Department of wood and forest sciences, Université Laval (2023-2026) - \$60,000

Master's degree - \$750

[2] Master's fellowship, University of Parakou, Benin Republic (2019-2020) - \$750

Bachelor's degree - \$2500

[1] Bachelor's scholarship, Minister of Higher Education and scientific research, Benin Republic (2014-2017) - \$2500

Professional membership

Society of Ecological restoration since 2024

British Ecological Society since 2019

Mentoring

Graduate co-advisor

[3] Franel Affolabi 2025. Sélection des espèces à racines traçantes dans le contexte agroforestier post minier. Master's degree, Université Laval, Québec, Canada.

[2] Justine Daviault 2025. Pratiques de revégétalisation des sites miniers générateurs de drainage minier acide au québec : acquis, opportunités et défis. Master's degree, Université Laval, Québec, Canada.

Undergraduate co-advisor

[1] Osé Zoundoh 2020. Effets de différentes doses d'amendements organique et minérale sur la croissance juvénile de Afzelia, khaya et Pterocarpus. Bachelor's degree, University of Parakou, Parakou, Benin